

Narsi Reddy Sanikommu

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EDUCATION

Master of Science - Mechanical Engineering

Aug 2024 - May 2026

JOHNS HOPKINS UNIVERSITY

Baltimore, MD

Bachelor of Technology - Mechanical Engineering

Jul 2016- May 2020

National Institute of Technology (NIT Trichy)

Tiruchirappalli, India

EXPERIENCE

Becton, Dickinson, And Company (BD), Bio-sciences

Jul 2025 - Jan 2026

Mechanical Engineering Co-op Intern

Sparks, MD, USA

- Designed and validated a shuttle fixture for the **BD COR GX shuttle**, enabling 20,000 cycle durability testing vs 500 cycles previously, a spring-based mechanism replicating tube interaction behavior, eliminating frequent tube replacement, significantly improving test efficiency and reliability.
- Built a 3D sensor response contour model for **BD COR MX** liquid waste detection sensor using **numPy sciPy**, achieving 95% prediction accuracy of trigger volume and supporting design optimization decisions.
- Performed feasibility study and design iteration of the **BD COR PX load cell** using **SolidWorks Flow simulation**, enabling real-time measurement of sample, diluent, and reagent weights, for improved error detection and spill prevention.
- Executed **RCA** on the **BD COR MX extractor** magnetic-strength characterization fixture; implemented minimal design changes and material changes with a snap-fit reinforcement, eliminating warping without triggering revalidation of test tool.

Dr Gracias Lab, JHU

Oct 2024 - Nov 2025

Research Assistant

Baltimore, MD

- Designed and fabricated soft, autonomous microneedle roll devices using **photolithography & soft lithography** for targeted drug delivery in the GI tract to treat autoimmune disorders.
- Optimized fabrication of drug-embedded gelatin microneedles (0.3 mm diameter) using soft lithography, reducing process time from **2 hours to 2 minutes** and improving mold-release yield from **40% to 80%**.

Becton, Dickinson, And Company (BD)

Jul 2020 - Aug 2023

Design Engineer 2, Design Engineer 1, GET

Bangalore, India

- Led multiple PE projects involving material change by updating **SolidWorks** drawings per **GD&T** standards, planned and executed DV activities, analyzed data with **Minitab**, DV Protocols & report release, DV Reviews, and ECRO process through **Team-center PLM** and **SAP**. Completed projects 2 weeks ahead of schedule, avoided back-orders and operation delays.
- Coordinated **Test Method updates** by designing a new test fixture and performed **MSA** and **FEA** simulations to replicate fixture engagement with components. Updated and leveraged existing MSA, reducing DV tests by **5x** and saving **2 weeks** of testing time.
- Updated and validated **112 BOMs in SAP**, preventing duplication errors and identifying 5 critical BOM issues; avoiding **3 weeks** of potential rework.
- Conducted **RCA** of high-pressure leak-test failures in **BD Alaris pump sets**; implemented fixture and work-instruction updates that eliminated re-testing, preserved **100%** of DV test data, and avoided an estimated **\$20K** in retest costs and schedule delays.
- Developed rapid prototype for new HDS product using existing BD PIVC platform components in **1 week at a cost of \$100**, accelerating concept validation and proposing an NPD timeline that cuts time-to-market by **6 months**.
- Contributed to innovation pipeline with **68 invention disclosures**, **14 US patent** filings, and multiple ongoing submissions.

NOTABLE PROJECTS

Segmented Fluidics Flow for Soft-body actuation, & Self folding shell microfluidics | Baltimore, MD Sep 2025 - Present

- Fabricated segmented fluidic-flow actuation in soft-bodied systems by developing a single-channel hydrogel (**PEGDA-AAm hydrogel**) "noodle" actuator; implemented segment-wise expansion control using timed segmented fluid delivery (water) and dehydration cycles (IPA) to achieve flexible robot motion without any external actuation mechanisms.
- Investigated a **multimodal, single-channel microfluidic platform** to route drugs and nutrients to multiple target sites; applied the approach to self-folding shell organoids and developed low-cost, reusable SLA-printed prototypes.

SKILLS

Programming	Python C++ C
CAD & Simulation	SolidWorks CATIA Creo NX AutoCAD Onshape ANSYS Abaqus
Tools	MATLAB Minitab ROS2 Arduino IDE KiCad Teamcenter PLM SAP
Fabrication	SLA FDM DLS BMF Photolithography Soft Lithography
Core Engineering	GD&T DFSS DFMA MSA RCA CAPA FEA FMEA
Research Areas	Microfluidics MEMS devices Cleanroom fabrication TEM/SEM characterization

NOTABLE PUBLICATIONS

Stable flow regulator assembly:	US20240175499A1 (Application grant)
Fluid connector system:	US12186518B2 (Application grant)
Comprehensive list and details of the patents:	(Google Scholar)